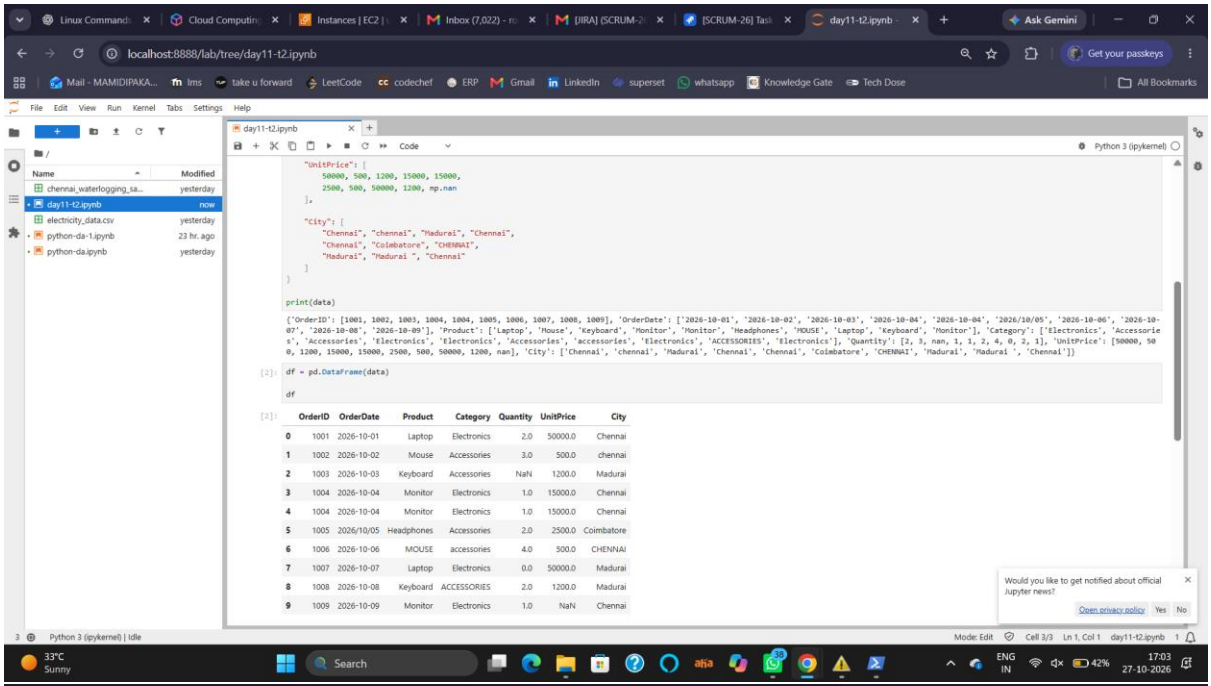


```
    "UnitPrice": [
        50000, 500, 1200, 15000, 15000,
        2500, 500, 50000, 1200, np.nan
    ],

    "City": [
        "Chennai", "chennai", "Madurai", "Chennai",
        "Chennai", "Coimbatore", "CHENNAI",
        "Madurai", "Madurai ", "Chennai"
    ]
}

print(data)
```

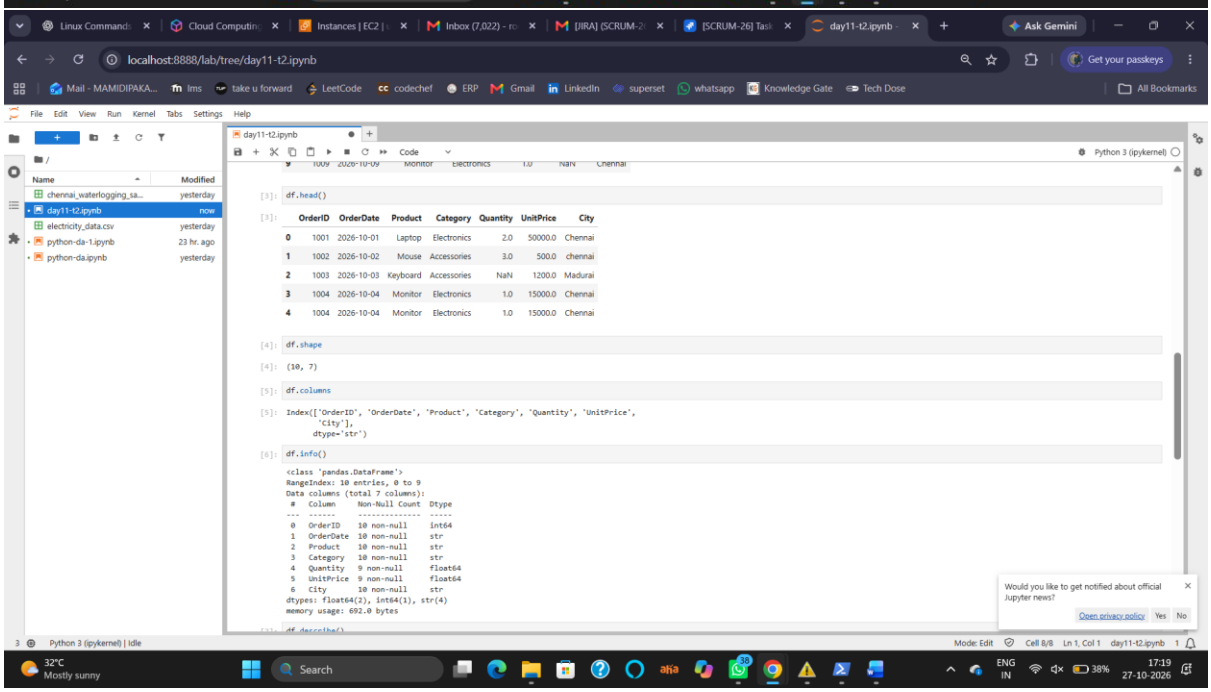
```
{'OrderID': [1001, 1002, 1003, 1004, 1004, 1005, 1006, 1007, 1008, 1009], 'OrderDate': ['2026-10-01', '2026-10-02', '2026-10-03', '2026-10-04', '2026-10-04', '2026/10/05', '2026-10-06', '2026-10-07', '2026-10-08', '2026-10-09'], 'Product': ['Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Monitor', 'Headphones', 'MOUSE', 'Laptop', 'Keyboard', 'Monitor'], 'Category': ['Electronics', 'Accessories', 'Accessories', 'Electronics', 'Electronics', 'Accessories', 'accessories', 'Electronics', 'ACCESSORIES', 'Electronics'], 'Quantity': [2, 3, nan, 1, 1, 2, 4, 0, 2, 1], 'UnitPrice': [50000, 500, 1200, 15000, 15000, 2500, 500, 50000, 1200, nan], 'City': ['Chennai', 'chennai', 'Madurai', 'Chennai', 'Chennai', 'Coimbatore', 'CHENNAI', 'Madurai', 'Madurai ', 'Chennai']}
```

```
[2]: df = pd.DataFrame(data)

     df
```

[2]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
0	1001	2026-10-01	Laptop	Electronics	2.0	50000.0	Chennai
1	1002	2026-10-02	Mouse	Accessories	3.0	500.0	chennai
2	1003	2026-10-03	Keyboard	Accessories	NaN	1200.0	Madurai
3	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai
4	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai
5	1005	2026/10/05	Headphones	Accessories	2.0	2500.0	Coimbatore
6	1006	2026-10-06	MOUSE	accessories	4.0	500.0	CHENNAI
7	1007	2026-10-07	Laptop	Electronics	0.0	50000.0	Madurai
8	1008	2026-10-08	Keyboard	ACCESSORIES	2.0	1200.0	Madurai
9	1009	2026-10-09	Monitor	Electronics	1.0	NaN	Chennai



```
[3]: df.head()
```

[3]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
0	1001	2026-10-01	Laptop	Electronics	2.0	50000.0	Chennai
1	1002	2026-10-02	Mouse	Accessories	3.0	500.0	chennai
2	1003	2026-10-03	Keyboard	Accessories	NaN	1200.0	Madurai
3	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai
4	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai

```
[4]: df.shape
```

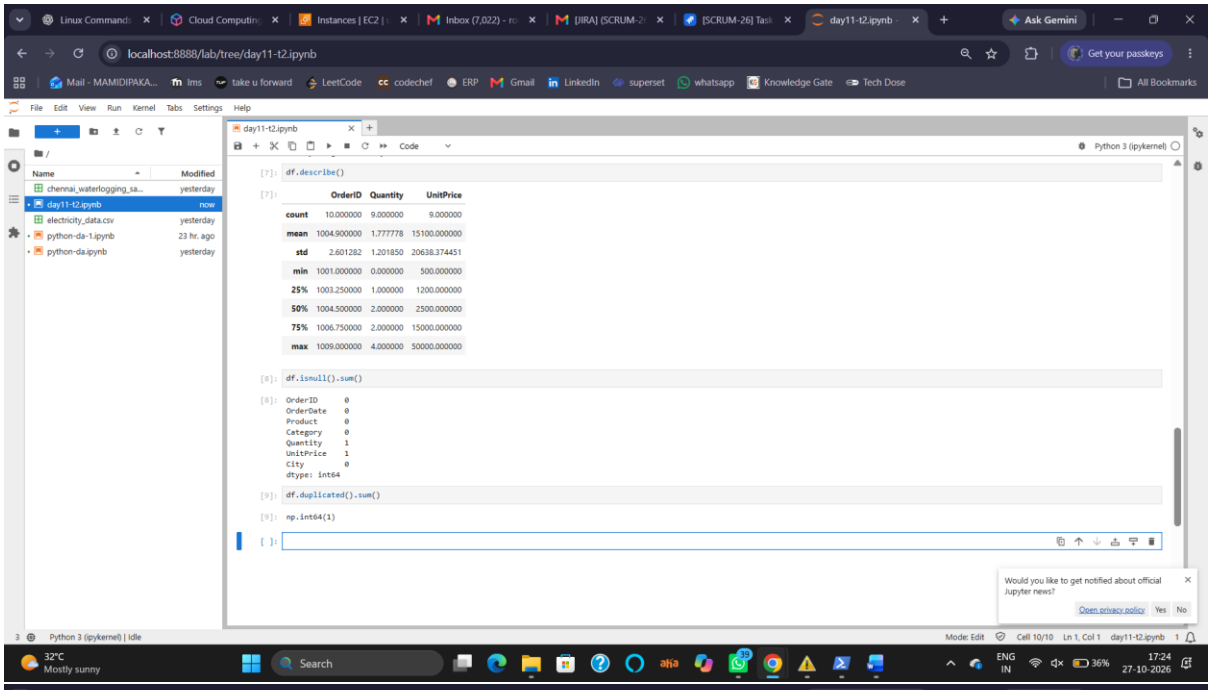
```
[4]: (10, 7)
```

```
[5]: df.columns
```

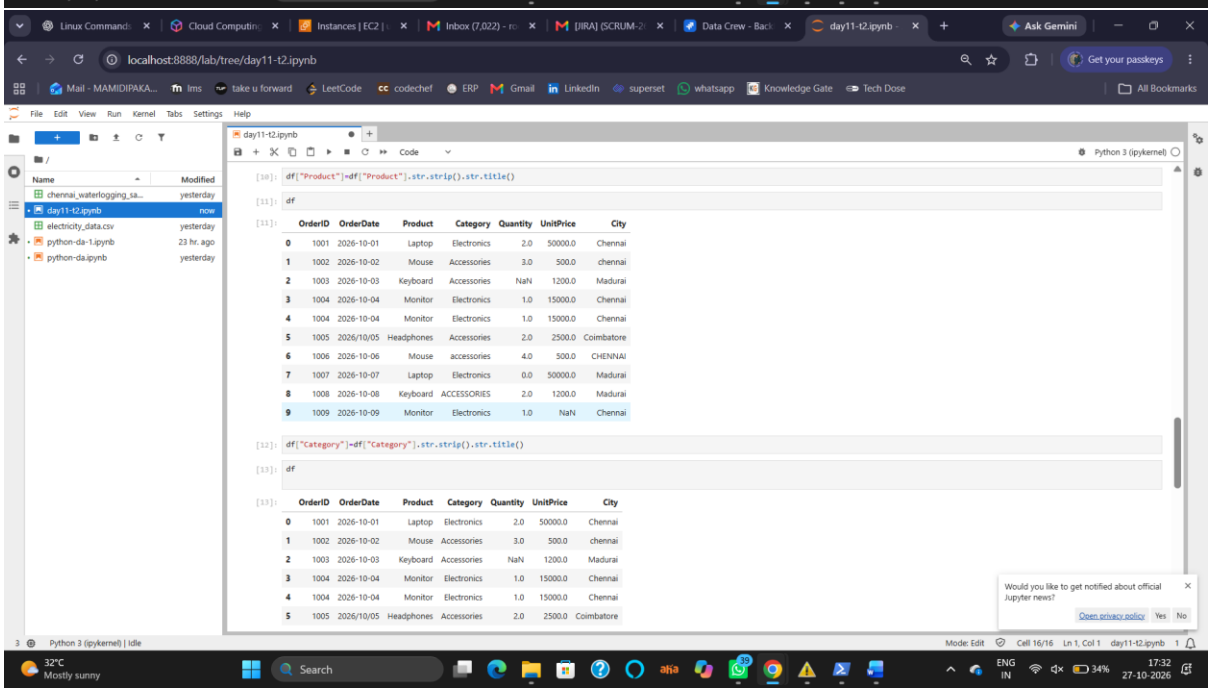
```
[5]: Index(['OrderID', 'OrderDate', 'Product', 'Category', 'Quantity', 'UnitPrice',
       'City'],
      dtype='str')
```

```
[6]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   OrderID    10 non-null     int64
 1   OrderDate  10 non-null     str
 2   Product    10 non-null     str
 3   Category   10 non-null     str
 4   Quantity   9 non-null      float64
 5   UnitPrice  9 non-null      float64
 6   City       10 non-null     str
dtypes: float64(2), int64(1), str(4)
memory usage: 692.0 bytes
```



	OrderID	Quantity	UnitPrice
count	10.000000	9.000000	9.000000
mean	1004.900000	1.777778	15100.000000
std	2.601282	1.201850	20638.374451
min	1001.000000	0.000000	500.000000
25%	1003.250000	1.000000	1200.000000
50%	1004.500000	2.000000	2500.000000
75%	1006.750000	2.000000	15000.000000
max	1009.000000	4.000000	50000.000000

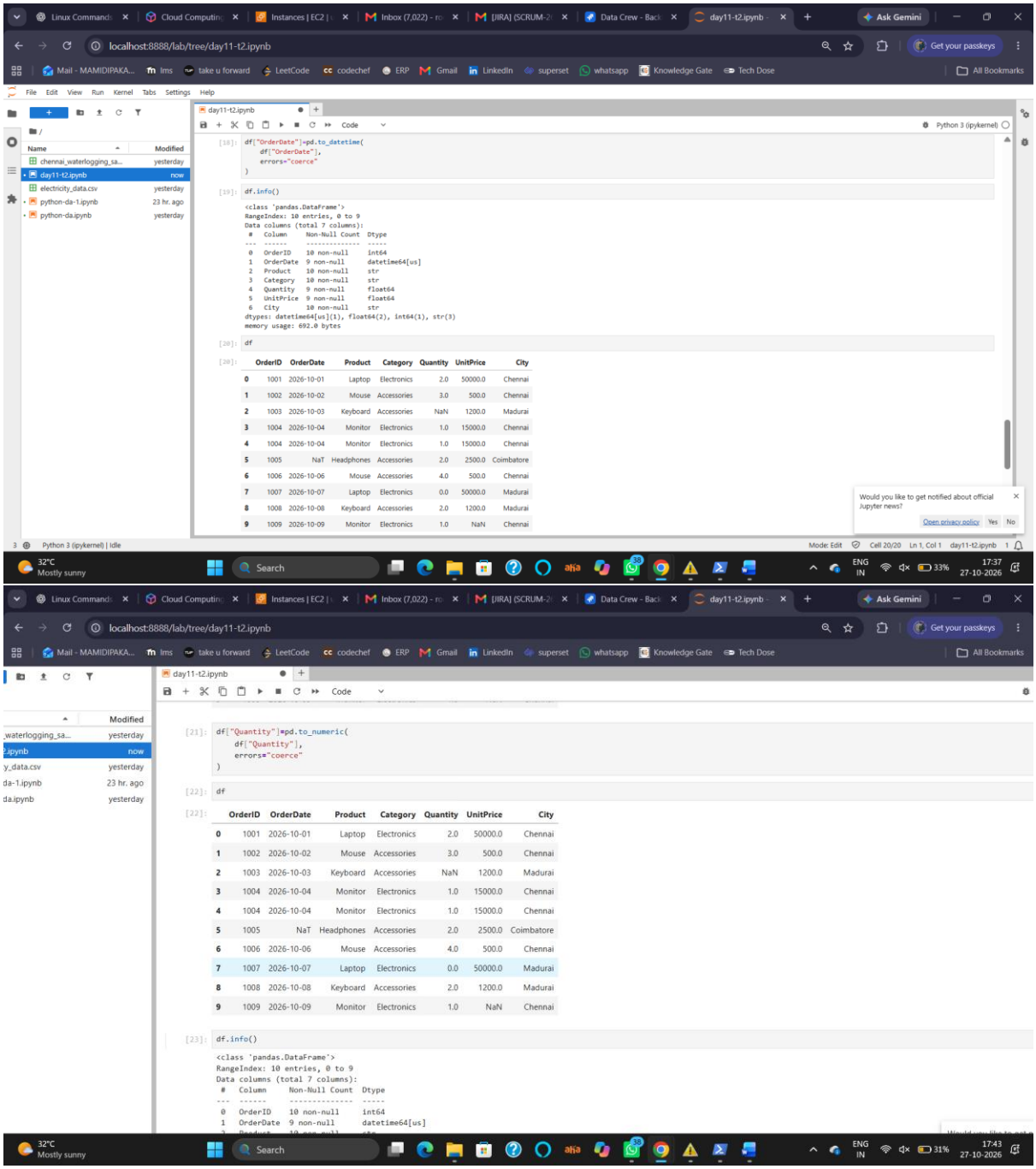


[11]:

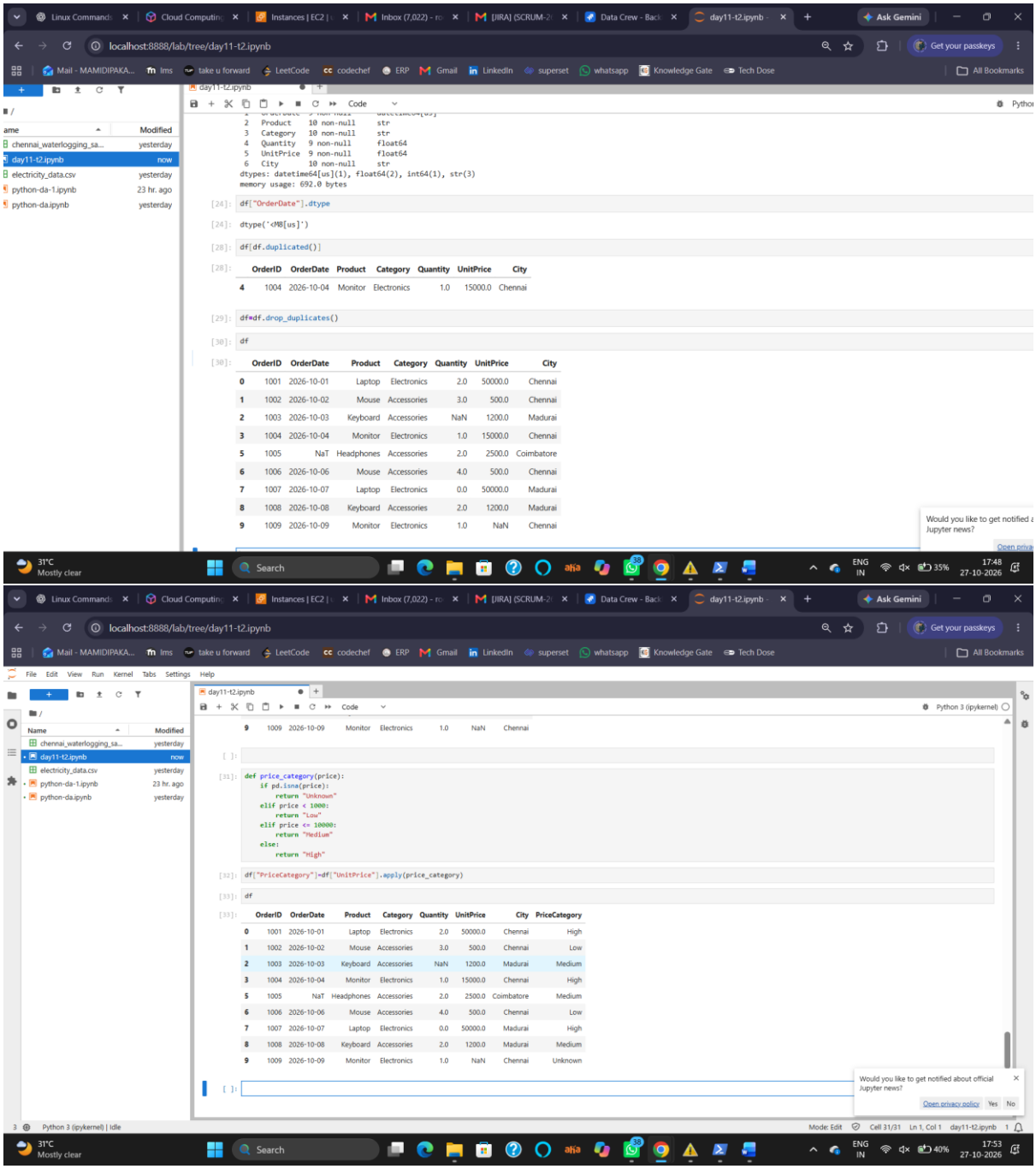
	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
0	1001	2026-10-01	Laptop	Electronics	2.0	50000.0	Chennai
1	1002	2026-10-02	Mouse	Accessories	3.0	500.0	chennai
2	1003	2026-10-03	Keyboard	Accessories	NaN	1200.0	Madurai
3	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai
4	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai
5	1005	2026/10/05	Headphones	Accessories	2.0	2500.0	Coimbatore
6	1006	2026-10-06	Mouse	accessories	4.0	500.0	CHENNAI
7	1007	2026-10-07	Laptop	Electronics	0.0	50000.0	Madurai
8	1008	2026-10-08	Keyboard	ACCESSORIES	2.0	1200.0	Madurai
9	1009	2026-10-09	Monitor	Electronics	1.0	NaN	Chennai

[13]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
0	1001	2026-10-01	Laptop	Electronics	2.0	50000.0	Chennai
1	1002	2026-10-02	Mouse	Accessories	3.0	500.0	chennai
2	1003	2026-10-03	Keyboard	Accessories	NaN	1200.0	Madurai
3	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai
4	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai
5	1005	2026/10/05	Headphones	Accessories	2.0	2500.0	Coimbatore


localhost:8888/lab/tree/day11-t2.ipynb
day11-t2.ipynb
Code
Python 3 (ipykernel)
[18]:
df["OrderDate"]=pd.to_datetime(
df["OrderDate"],
errors="coerce"
)
[19]:
df.info()
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
Column Non-Null Count Dtype
--- ------ -------------- -----
0 OrderID 10 non-null int64
1 OrderDate 9 non-null datetime64[us]
2 Product 10 non-null str
3 Category 10 non-null str
4 Quantity 9 non-null float64
5 UnitPrice 9 non-null float64
6 City 10 non-null str
dtypes: datetime64[us](1), float64(2), int64(1), str(3)
memory usage: 692.0 bytes
[20]:
df
[20]:
OrderID OrderDate Product Category Quantity UnitPrice City
0 1001 2026-10-01 Laptop Electronics 2.0 50000.0 Chennai
1 1002 2026-10-02 Mouse Accessories 3.0 500.0 Chennai
2 1003 2026-10-03 Keyboard Accessories NaN 1200.0 Madurai
3 1004 2026-10-04 Monitor Electronics 1.0 15000.0 Chennai
4 1004 2026-10-04 Monitor Electronics 1.0 15000.0 Chennai
5 1005 NaT Headphones Accessories 2.0 2500.0 Coimbatore
6 1006 2026-10-06 Mouse Accessories 4.0 500.0 Chennai
7 1007 2026-10-07 Laptop Electronics 0.0 50000.0 Madurai
8 1008 2026-10-08 Keyboard Accessories 2.0 1200.0 Madurai
9 1009 2026-10-09 Monitor Electronics 1.0 NaN Chennai
Would you like to get notified about official Jupyter news?
Open privacy policy
Yes
No
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 20/20
Ln 1, Col 1
day11-t2.ipynb
32°C
Mostly sunny
Search
ENG
IN
33%
17:37
27-10-2026

localhost:8888/lab/tree/day11-t2.ipynb
day11-t2.ipynb
Code
[21]:
df["Quantity"]=pd.to_numeric(
df["Quantity"],
errors="coerce"
)
[22]:
df
[22]:
OrderID OrderDate Product Category Quantity UnitPrice City
0 1001 2026-10-01 Laptop Electronics 2.0 50000.0 Chennai
1 1002 2026-10-02 Mouse Accessories 3.0 500.0 Chennai
2 1003 2026-10-03 Keyboard Accessories NaN 1200.0 Madurai
3 1004 2026-10-04 Monitor Electronics 1.0 15000.0 Chennai
4 1004 2026-10-04 Monitor Electronics 1.0 15000.0 Chennai
5 1005 NaT Headphones Accessories 2.0 2500.0 Coimbatore
6 1006 2026-10-06 Mouse Accessories 4.0 500.0 Chennai
7 1007 2026-10-07 Laptop Electronics 0.0 50000.0 Madurai
8 1008 2026-10-08 Keyboard Accessories 2.0 1200.0 Madurai
9 1009 2026-10-09 Monitor Electronics 1.0 NaN Chennai
[23]:
df.info()
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
Column Non-Null Count Dtype
--- ------ -------------- -----
0 OrderID 10 non-null int64
1 OrderDate 9 non-null datetime64[us]
32°C
Mostly sunny
Search
ENG
IN
31%
17:43
27-10-2026



```
 2   Product     10 non-null     str
 3   Category    10 non-null     str
 4   Quantity    9 non-null      float64
 5   UnitPrice   9 non-null      float64
 6   City        10 non-null     str
dtypes: datetime64[us](1), float64(2), int64(1), str(3)
memory usage: 692.0 bytes
```

```
[24]: df["OrderDate"].dtype
```

```
[24]: dtype('<M8[us]')
```

```
[28]: df[df.duplicated()]
```

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
4	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai

```
[29]: df=df.drop_duplicates()
```

```
[30]: df
```

[30]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
0	1001	2026-10-01	Laptop	Electronics	2.0	50000.0	Chennai
1	1002	2026-10-02	Mouse	Accessories	3.0	500.0	Chennai
2	1003	2026-10-03	Keyboard	Accessories	NaN	1200.0	Madurai
3	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai
5	1005	NaT	Headphones	Accessories	2.0	2500.0	Coimbatore
6	1006	2026-10-06	Mouse	Accessories	4.0	500.0	Chennai
7	1007	2026-10-07	Laptop	Electronics	0.0	50000.0	Madurai
8	1008	2026-10-08	Keyboard	Accessories	2.0	1200.0	Madurai
9	1009	2026-10-09	Monitor	Electronics	1.0	NaN	Chennai

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```
 9   1009   2026-10-09   Monitor   Electronics   1.0   NaN   Chennai
```

```
[ ]:
```

```
[31]: def price_category(price):
          if pd.isna(price):
              return "Unknown"
          elif price < 1000:
              return "Low"
          elif price <= 10000:
              return "Medium"
          else:
              return "High"
```

```
[32]: df["PriceCategory"]=df["UnitPrice"].apply(price_category)
```

```
[33]: df
```

[33]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City	PriceCategory
0	1001	2026-10-01	Laptop	Electronics	2.0	50000.0	Chennai	High
1	1002	2026-10-02	Mouse	Accessories	3.0	500.0	Chennai	Low
2	1003	2026-10-03	Keyboard	Accessories	NaN	1200.0	Madurai	Medium
3	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai	High
5	1005	NaT	Headphones	Accessories	2.0	2500.0	Coimbatore	Medium
6	1006	2026-10-06	Mouse	Accessories	4.0	500.0	Chennai	Low
7	1007	2026-10-07	Laptop	Electronics	0.0	50000.0	Madurai	High
8	1008	2026-10-08	Keyboard	Accessories	2.0	1200.0	Madurai	Medium
9	1009	2026-10-09	Monitor	Electronics	1.0	NaN	Chennai	Unknown

```
[ ]:
```

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